Techiniques, Examples And Counterexamples

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- **Modules** $(M, R, +)$

  Proof. Verify the axioms. ☐

- **Submodules**

  Proof. Verify the axioms (Check subgroup and for any $r \in R$, $m \in M$, $rm \in M$). ☐

- **Homomorphism**

  Proof. Verify the axioms. ☐

Example 0.1. 1. Vector space $\mathbb{C}$ (Ring over module is a field).

2. (V, T) where V is a vector space and T is a linear transform. Ring is constructed to be $F[x]$ where

$$f(x) \cdot v := f(T)(v).$$

Note: $(V, T) \longmapsto F[x]$ module. For $\longmapsto$ we define $T(x) := x \cdot v$.

- **Vector subspaces**
- In $\mathbb{Z}$-modules say M

  Definition 0.1. N is a submodule iff $(N, +) \leq (CM, +)$.

- In R is a R-module

  Definition 0.2. N is a submodule iff N is an ideal.

- If M is a $F[x]$-module $\equiv (V, T)$

  Definition 0.3. W is a submodule iff $T(CW) \subseteq W$, i.e., W is invariant under T.

Finitely generated and cyclic modules

Module	**f.g.**	**cyclic**
$(V, \mathbb{F})$	$\dim V < \infty$	$\dim V = \{0, 1\}$
G is abelian group treated like $\mathbb{Z}$-module	f.g as group	cyclic as group
R comm ring and $M \triangleleft R$ M is R-module	f.g as an ideal	principle ideal

Definition 0.4. A module M is *finitely generated* if there exists a finite set of elements $\{m_1, m_2, \ldots, m_n\} \subseteq M$ such that every element of M can be written as a linear combination of these elements with coefficients from the ring R.

Definition 0.5. A module M is *cyclic* if it is generated by a single element, i.e., there exists an element $m \in M$ such that every element of M can be written as rm for some r in the ring R.

Remark 0.1. For a vector space $(V, \mathbb{F})$ over a field $\mathbb{F}$, the module V is finitely generated if and only if its dimension is finite, i.e., $\dim V < \infty$. It is cyclic if and only if $\dim V \leq 1$.

Remark 0.2. An abelian group G treated as a $\mathbb{Z}$-module is finitely generated as a group if and only if it is finitely generated as a $\mathbb{Z}$-module. It is cyclic as a group if and only if it is cyclic as a $\mathbb{Z}$-module.

Remark 0.3. If R is a commutative ring and M is an R-module, then M is finitely generated as an R-module if it is finitely generated as an ideal. If M is a principal ideal, then it is cyclic as an R-module.

- **Module homomorphisms**
  - Abelian groups $\leftrightarrow$ $\mathbb{Z}$-modules, then group homomorphisms $\leftrightarrow$ module homomorphisms.
  - For vector fields, linear transformations $\leftrightarrow$ module homomorphisms.
  - For a ring R, $\Theta : M \to N$ where $M = N = R$ and

    $$\Theta(rs) = r\Theta(s)$$

    is **not** a ring homomorphism but is a module homomorphism.
  - Isomorphism theorems hold for modules.